Abdelrahman Elmogy

Data & Automation Engineer

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Cairo, Egypt · Open to Relocation across Europe

PROFESSIONAL SUMMARY

Data & Automation Engineer with 3+ years of experience building data pipelines, integration workflows, reporting systems, and business automation solutions. Strong hands-on experience with Python, SQL Server, ETL, REST APIs, Power BI, and operational databases. Works from the business problem outward: understands the process, identifies the root cause, designs the right data solution, then builds, integrates, and automates it end to end.

7+ Systems	60%	8 hrs/week	10+
Automated	Manual Effort Reduced	Reporting Time Saved	High-Impact Projects

TECHNICAL SKILLS

Data Engineering	Python · Pandas · NumPy · ETL · Data Pipelines · Database Design · SQLAlchemy · pyodbc
System Integration	REST APIs · API Integration · Webhooks · Data Integration · SQL Databases · Real-Time Data Flows
BI & Reporting	Power BI · DAX · Power Query · SQL Server · Excel · KPI Design · Dashboard Architecture
Automation & AI	Workflow Automation · Business Process Automation · AI Agents · Conversational AI · Task Scheduling
Development	Flask · FastAPI · Selenium · BeautifulSoup · openpyxl · Git
Infrastructure	Windows Server · Linux · cPanel · System Migration

PROFESSIONAL EXPERIENCE

Data & Automation Engineer · Maharah International for Debt Collection

Jul 2024 – Sep 2026

Cairo, Egypt

Data Integration & AI-Powered Automation

► Analysed operational bottlenecks in a high-volume outbound process, then designed and built an AI-powered conversational solution integrated with SQL Server, reading and writing live operational data in real time. First solution of its kind at the company.

Business Intelligence & Reporting

► Analysed reporting gaps across departments and built live Power BI dashboards connected directly to SQL Server, replacing a manual weekly reporting process and giving management real-time, production-data visibility. Saved approximately 8 hours per week.

Data Pipelines & Workflow Automation

► Identified recurring data-entry errors through workflow analysis, then designed and built Python-based automation pipelines that reduced manual processing effort by 60% and eliminated discrepancies affecting financial reporting.

► Studied multi-channel dispatch patterns and engineered a data pipeline pulling segmented records from SQL Server and distributing them across IVR, WhatsApp, and email channels simultaneously, writing outcomes back automatically. Cut execution time by 60%.

Operational Analytics & Automation Strategy

► Led data-driven performance analysis across multiple communication channels using Power BI and Python, translating findings into optimisation and automation strategies adopted across the department.

System Administrator & Data Analyst · Royal Canadian School

Aug 2023 – Jun 2024

New Administrative Capital, Egypt

► Analysed fragmented record-keeping and built data management systems in Excel and SQL Server covering 500+ student and staff records.

► Studied the finance department's forecasting workflow and designed a live Power BI budget tracking dashboard, enabling accurate real-time financial forecasting for the first time.
► Managed full IT infrastructure, including networks, servers, and workstations, maintaining high uptime across multiple departments.

Freelance Python Developer · Independent · Remote **Oct 2023 – Present**

► Analysed client workflows across multiple industries and built end-to-end Python automation systems, including web scraping pipelines, WhatsApp bots, and email campaign engines, reducing client workloads by up to 40% per engagement.
► Studied reporting needs for clients in finance, retail, and services, then developed custom data analysis solutions that replaced manual spreadsheets with automated dashboards for faster decision-making.
► Assessed operational gaps and designed business automation tools, including invoice processors, reconciliation scripts, and scheduled reporting systems, integrating SQL databases and REST APIs for reliable client solutions.

SELECTED PROJECTS

01 Maharah AI Collection Agent · *Conversational AI · Python · SQL Server · Real-Time API Integration*
Designed a real-time data integration layer connecting a conversational AI engine to SQL Server, enabling live account data to flow into and out of active conversations. Built the solution end to end, from data access logic to AI response handling.

02 Fraud Detection Model · *Python · Scikit-learn · Pandas · Machine Learning*
Built a data pipeline processing transaction-level records to extract device-signature clusters, timing anomalies, and usage outliers. Developed the underlying model that scores and ranks transactions into a prioritised risk queue, replacing manual review.

03 Cross Channel Campaign Dispatch Engine · *Python · SQL Server · API Integration · Multi-Channel Automation*
Engineered a data pipeline pulling segmented records from SQL Server and integrating with IVR, WhatsApp, and email APIs to dispatch campaigns simultaneously, with outcomes written back automatically. Cut execution time by 60%.

04 Maharah Collection Performance Tracker · *Power BI · DAX · SQL Server · Real-Time Dashboards*
Mapped reporting requirements across two organisations, then built a Power BI reporting layer connected directly to live SQL Server data, replacing static exports with real-time operational and financial dashboards.

05 Automated Commission Calculation Engine · *Python · Excel Automation · Data Processing*
Diagnosed a slow, error-prone manual Excel process as the root cause of recurring payroll mistakes. Built an automated data-processing pipeline that recalculates commissions every cycle without manual input, removing calculation errors entirely.

06 TalkHub, a WhatsApp Business Automation System · *Python · WhatsApp API · Messaging Automation*
Integrated with the WhatsApp Business API to build a full-featured messaging platform supporting two-way conversations, automated replies, broadcast campaigns, and complete conversation logging for business clients.

07 LegalDesk, a Case and Document Management Platform · *Python · SQL · Document Automation*
Mapped a client's intake and documentation workflow end to end, then built a database-backed platform automating client intake, contract generation, case tracking, and document management.

08 PeopleOps, an HR Operations Platform · *Python · SQL · Process Automation*
Audited workflow gaps across employee records, attendance, and payroll, then built a connected database platform integrating attendance tracking, recruitment, employee records, and payroll processing.

EDUCATION

Bachelor of Communication Engineering
Mansoura University, Egypt · 2018–2023

LANGUAGES

Arabic Native, **English** Prof, **Italian** Elem

AVAILABILITY

Available Immediately
Open to EU Relocation
On-site / Hybrid / Remote

CERTIFICATIONS

Data Analysis Professional, *Udacity 2023*
Data Analysis Challenger, *Udacity 2022*
Excel + Power BI Diploma, *Planet of Skills 2023*
Python & SQL Advanced, *HackerRank 2022*
Cyber Ops Associate, *NTI 2022*
CCNA Network Associate, *ITI 2021*
PL-300 Power BI Analyst, *Microsoft (In Progress)*